

Submission Info

- [Slides](#)
- Github: <https://github.com/yvonneanang/BME-Hacking-Health-2024>



PharMany

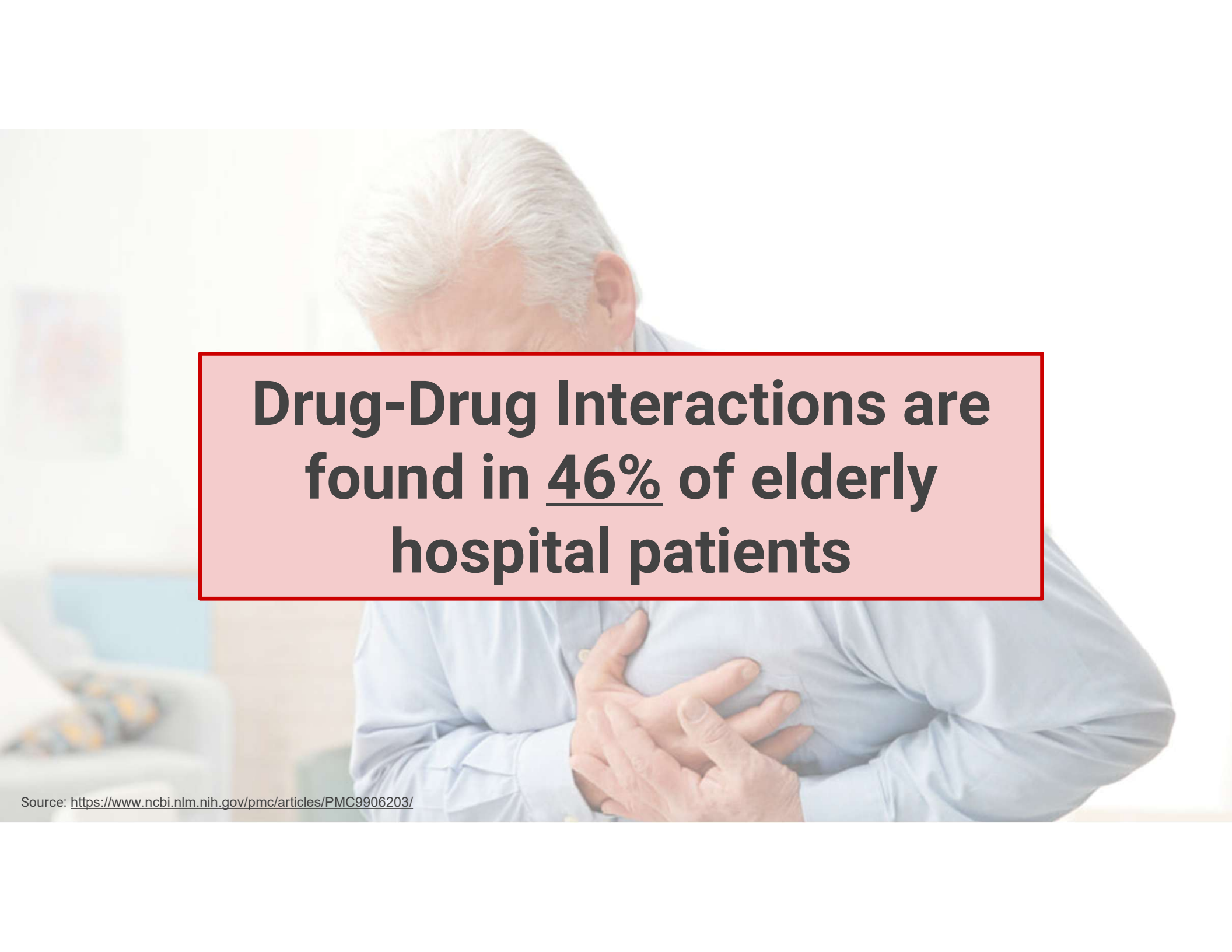
BME Healthcare Hackathon 2024





**60M Adults in the US are
taking 3 or more drugs**

Source: <https://www.cdc.gov/nchs/fastats/drug-use-therapeutic.htm>

A photograph of an elderly man with white hair, wearing a light blue button-down shirt. He is holding his right hand over his chest, suggesting discomfort or pain. The background is a blurred indoor setting, possibly a home or a hospital room.

**Drug-Drug Interactions are
found in 46% of elderly
hospital patients**

Source: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9906203/>

**Polypharmacy complications
is a \$62 Billion problem**

Source: <https://lowinstitute.org/projects/medication-overload-how-the-drive-to-prescribe-is-harming-older-americans/>



Why do these dangerous prescriptions happen?

It's a complicated mess

----- DRUG INTERACTIONS -----

- Strong cytochrome P450 3A4 inhibitors (e.g., ritonavir, ketoconazole): Use not recommended. May increase risk of systemic corticosteroid and cardiovascular effects. (7.1)
- Monoamine oxidase inhibitors and tricyclic antidepressants: Use with extreme caution. May potentiate effect of salmeterol on vascular system. (7.2)
- Beta-blockers: Use with caution. May block bronchodilatory effects of beta-agonists and produce severe bronchospasm. (7.3)
- Diuretics: Use with caution. Electrocardiographic changes and/or hypokalemia associated with non-potassium-sparing diuretics may worsen with concomitant beta-agonists. (7.4)



It's a complicated mess

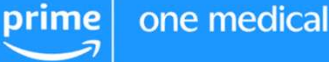
[Uses](#)[Side Effects](#)[Precautions](#)[Interactions](#)[Overdose](#)[Images](#)[Reviews \(1\)](#)

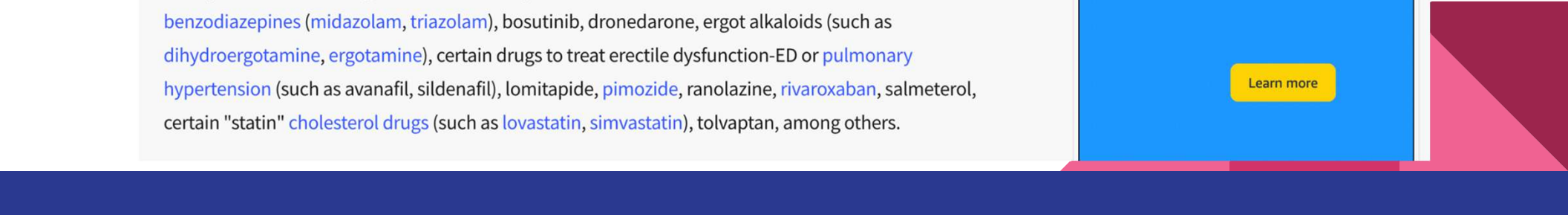
Interactions

[Drug interactions](#) may change how your [medications](#) work or increase your risk for serious side effects. This document does not contain all possible drug interactions. Keep a list of all the products you use (including prescription/nonprescription drugs and herbal products) and share it with your doctor and [pharmacist](#). Do not start, stop, or change the dosage of any medicines without your doctor's approval.

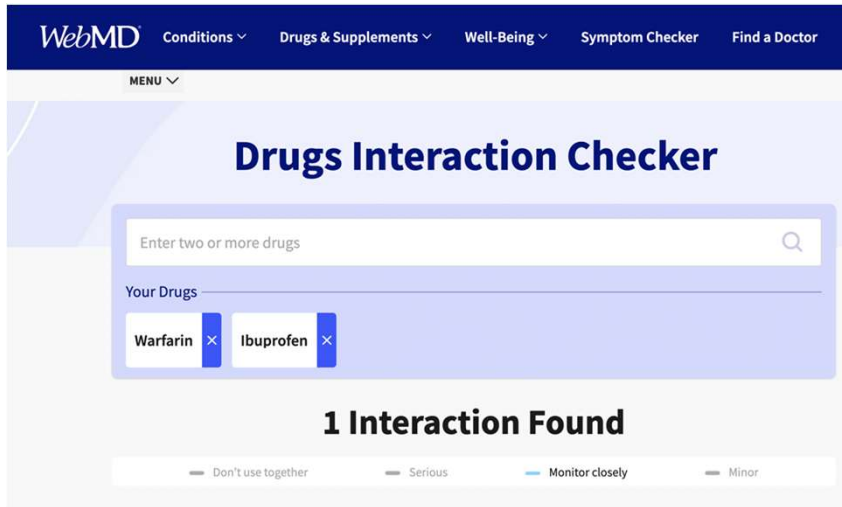
Some of the products that may interact with this drug include: [orlistat](#), certain drugs used to treat [seizures](#) (such as [carbamazepine](#), phenytoin, [primidone](#)), [barbiturates](#) (such as [phenobarbital](#), secobarbital), a certain combination [HIV medication](#) (elvitegravir/cobicistat/[emtricitabine](#)/[tenofovir](#)).

[Darunavir](#) can slow down the removal of other drugs from your body, which may affect how they work. Examples of affected drugs include certain alpha blockers (such as [alfuzosin](#), [tamsulosin](#)), certain [benzodiazepines](#) ([midazolam](#), [triazolam](#)), bosutinib, dronedarone, ergot alkaloids (such as [dihydroergotamine](#), [ergotamine](#)), certain drugs to treat erectile dysfunction-ED or [pulmonary hypertension](#) (such as avanafil, sildenafil), lomitapide, [pimozide](#), ranolazine, [rivaroxaban](#), salmeterol, certain "statin" [cholesterol drugs](#) (such as [lovastatin](#), [simvastatin](#)), tolvaptan, among others.

[Learn more](#)



There are some existing solutions...



The WebMD Drugs Interaction Checker interface features a dark blue header with navigation links: WebMD, Conditions, Drugs & Supplements, Well-Being, Symptom Checker, and Find a Doctor. Below the header is a 'MENU' dropdown. The main heading is 'Drugs Interaction Checker'. A search bar prompts 'Enter two or more drugs'. Below the search bar, 'Your Drugs' are listed: Warfarin and Ibuprofen. The results section shows '1 Interaction Found' with a legend indicating the severity: Don't use together, Serious, Monitor closely, and Minor.

FDA Label Search

FDA Home

Search by Active Ingredient:

(Type in part or all of active ingredient)

Clear Fields

Submit Query

[Return to the FDA Label Search Page](#)

...but they aren't

- personalized to a particular patient's prescriptions
- streamlined for medical professionals (multiple variables to consider)
- citing sources for further reference



PharMany!

Pharmaceutical assistant providing personalized, trustworthy, and in-depth reports for drug-drug interactions



How it works



Doctor enters
patient's new and
existing
prescriptions



How it works



Doctor enters
patient's new and
existing
prescriptions



PharMany searches **up-to-date FDA data sheets** of each drug to check for interacting active ingredients



How it works



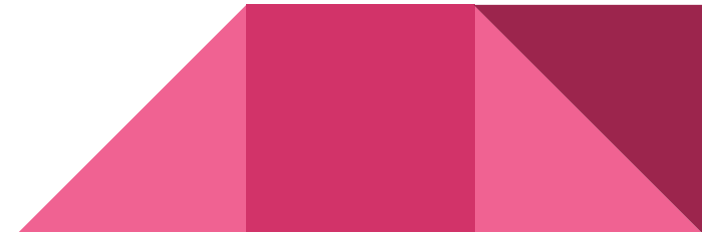
Doctor enters
patient's new and
existing
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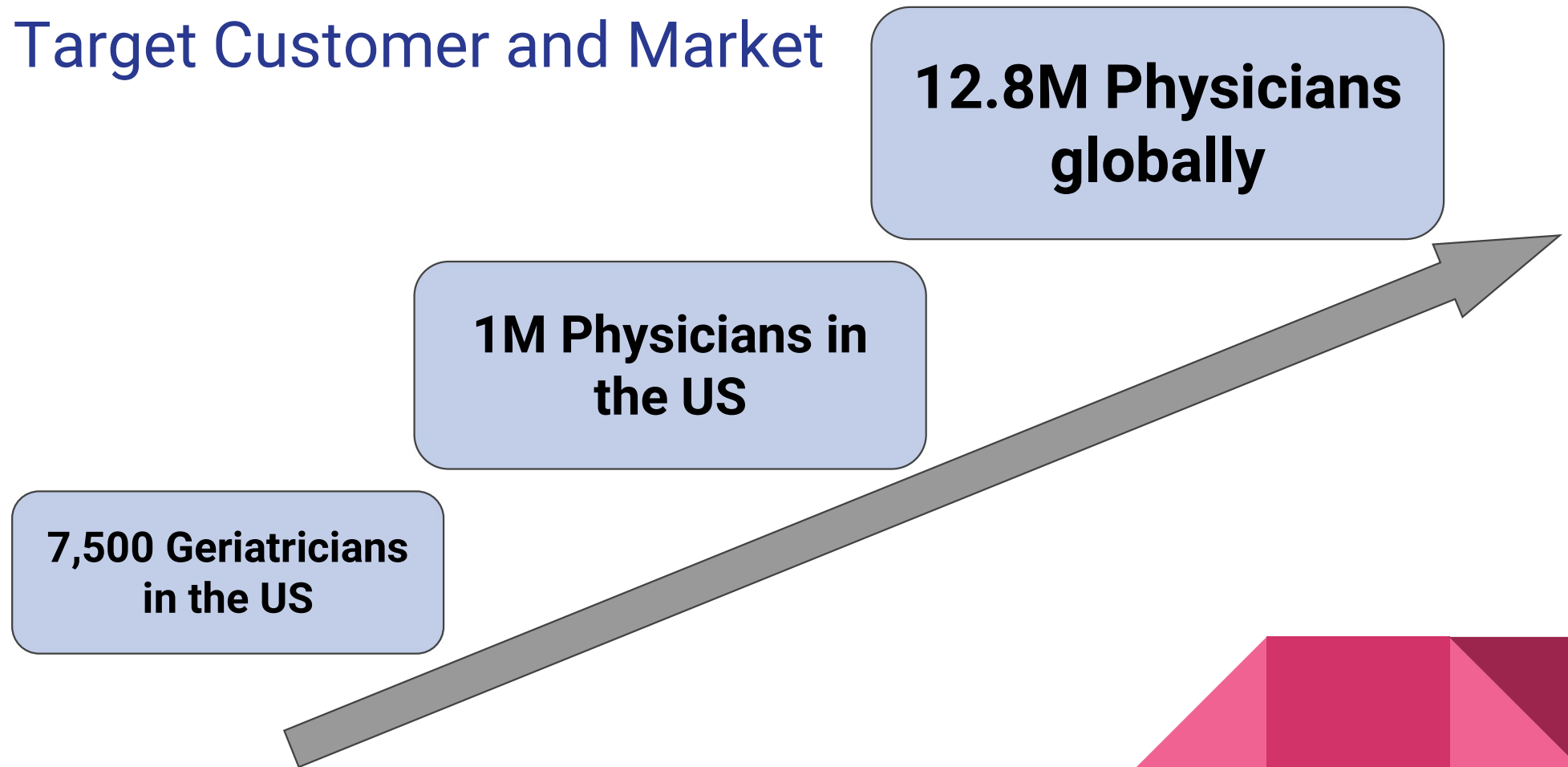
PharMany searches **up-to-date FDA data sheets** of each drug to check for interacting active ingredients



PharMany highlights **all predicted complications with citations**



Target Customer and Market



Source: [UNC Medicine](#), [Statista](#), [MedicalEconomics](#)

Demo

Implementation:

- Web Application

Tools used:

- Datasets from FDA API



Dashboard

Patients

Patient Portfolio

Drug Query

Drug Report

Settings

Patients



Audrey
Female



Emma
Female



Kevin
Male



Yvonne
Female

+

More
New User



Emma
Patient 2

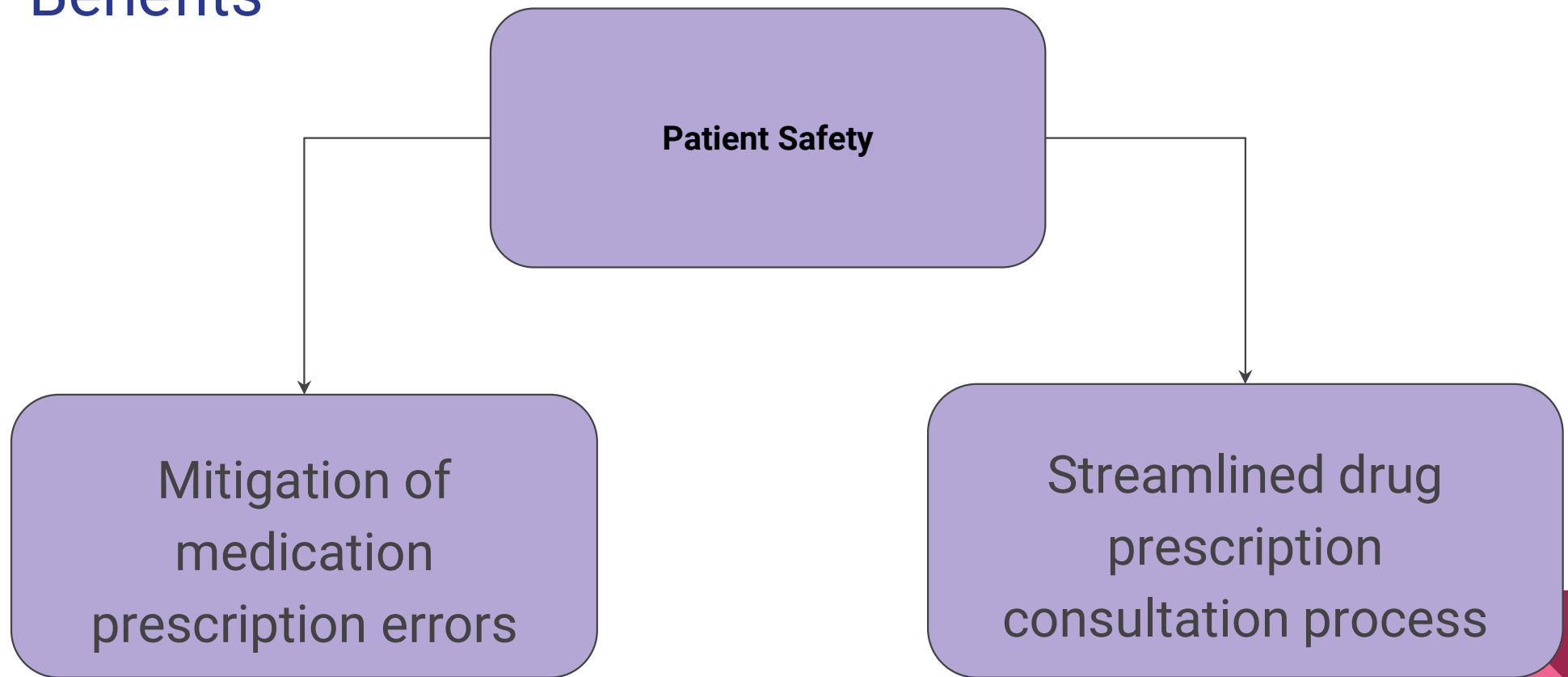
Drug Query

Query A Drug...

Drug Report

Drug Name	Queried Drug Active Ingredients	Patient Drug Active Ingredients	Patient Drug Portfliio
Mechanism of Action	Active Ingredients in queried drug	Active Ingredients for each drug in patient portfolio	
Bring me water to drink oh drug. I thirst for rejuvenation. Don't mix me with anything, thank you.	Active Ingredient 1	Active Ingredient 1	Drug 1
	Active Ingredient 2	Active Ingredient 2	Drug 2
	Active Ingredient 3	Active Ingredient 3	Drug 3
			Add Drug

Benefits



Next steps

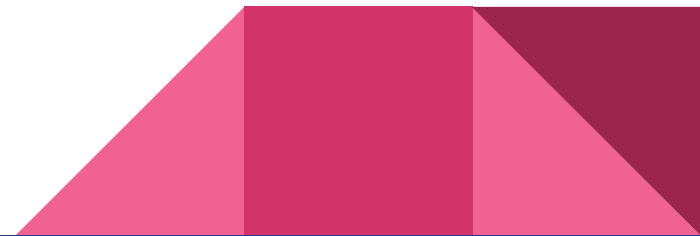
- More personalization: allergies, demographics with higher risk
- Add in information from adverse events database
- Use LLMs to incorporate the mechanism of action of a drug into consideration



Team



Questions?



Contact Info

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Background

- Background should contain your problem definition & key statistics to make us care about your problem
- Problem Definition:
 - Polypharmacy <https://www.cdc.gov/nchs/products/databriefs/db347.htm>
 - Medical error that can arise during the prescription of multiple drugs which have toxic drug-drug interactions
 - Paper, the effects of multiple drug prescriptions on patient health - <https://lowinstitute.org/projects/medication-overload-how-the-drive-to-prescribe-is-harming-older-americans/>
 - Therapeutic drug use CDC data - <https://www.cdc.gov/nchs/fastats/drug-use-therapeutic.htm>



Need Statement

- *Your need statement should be a complete sentence.*
- Visualize the cross reactivity of multiple drugs a patient might be taking with a new prescription a doctor might be adding to the patient's drug portfolio and outline possible interactions and adverse effects.
- This would help set streamline the process of identifying cross-reactivity across different drugs for patients such that a combination of drugs taken by a patient doesn't become lethal to the patient.



Existing Solutions/Prior Art

- FDALabel
- WebMD

Pros:

- Comprehensive database of approved drugs and their datasheets

Cons:

- Not personalized and specific to a particular patient's drug portfolio
- User experience is not favorable
- Data queries are not streamlined for medical professionals (multiple variables to consider)



Market Analysis

- TAM-SAM-SOM
 - *Be sure to report the total available market (TAM), serviceable available market (SAM), and serviceable obtainable market (SOM) in terms of dollar amounts (not number of people)*
- TAM - Percentage of patients who take multiple medicines
- SAM - Percentage of doctors prescribing multiple drugs per patient (and the cost per patient)
= (%doctors*number of patients *cost per patient)
- SOM - 20% market penetration (20%*number of doctors*number of patients*cost per patient)



Functional Requirements

- Personalized Patient Drug Portfolio
- Generated drug-drug interaction history report for medical professional use
- Reduction of medical error, i.e. mitigation of oversight



Application Demo

- Provide information about your prototyping journey & your final device



References

- Please provide long-form references here
 - *Remember that short-form references (i.e. Heckelman et al. (2022) or Heckelman⁺ (2022)) should be on the individual slide(s) as well*

